Stat 266 – Unit 6 – Video Set 1 – Class Examples

Summary:

Method of Distribution Functions

Note: We will only consider continuous random variables for this method.

Consider $U = h(Y_1, \ldots, Y_n)$ to be a function of the continuous random variables $Y_1, \ldots, Y_n$, which have joint density $f(y_1, \ldots, y_n)$.

- Find the region in the $(y_1, \ldots, y_n)$ space where $U \le u$.

- Find $F_U(u) = P(U \le u) = P(h(Y_1, \ldots, Y_n) \le u) = \int \cdots \int_{U \le u} f(y_1, \ldots, y_n) dy_1 \cdots dy_n$.

- Then, the density of U is $f_U(u) = \dfrac{d}{du} F_U(u)$.

Example A:

The amount of time waiting to be served, Y, is described by the following density

$$f(y)=\begin{cases}\frac{1}{\theta}e^{-y/\theta} & , \quad y>0 \\ 0 & , \quad else\end{cases}.$$

a) What is the probability distribution of $U=aY+b$, where $a>0$?

$$F_U(u)=P(U<u)=P(aY+b<u)=P\left(Y<\frac{u-b}{a}\right)$$

Now, we dont know if u is below 0 or above 0.

$$F_U(u)=\begin{cases}\int_0^{\frac{u-b}{a}}\frac{1}{\theta}e^{-y/\theta} & , \quad \frac{u-b}{a}>0 \\ 0 & , \quad \frac{u-b}{a}<0.\end{cases}$$

Case 1

$$\int_0^{\frac{u-b}{a}}\frac{1}{\theta}e^{-y/\theta}=\frac{1}{\theta}\cdot\frac{e^{-y/\theta}}{\frac{1}{\theta}\cdot -1}=\left[-e^{-y/\theta}\right]_0^{\frac{u-b}{a}}=-e^{\frac{-u+b}{a\theta}}-e^{0}=1-e^{\frac{-u+b}{a\theta}}$$

$$f_U(u)=\frac{d}{du}\,1-e^{\frac{-u+b}{a\theta}}=0-e^{\frac{-u+b}{a\theta}}\cdot\frac{1}{a\theta}\cdot(-1)=\frac{e^{\frac{-u+b}{a\theta}}}{a\theta}$$

b) What is the probability distribution of $U = Y^a$, where $a > 0$?

$$P(U < u) = P(Y^a < u) = P(Y < u^{1/a})$$

<u>Case 1</u>

$u < 0$

$$F_U(u) = \int_{-\infty}^{u^{1/a}} f_Y(y)\, dy = 0$$

<u>Case 2</u>

$u > 0$

$$F_U(u) = \int_{-\infty}^{u^{1/a}} f_Y(y)\, dy = \int_0^{u^{1/a}} \frac{1}{\theta} e^{-y/\theta}\, dy = \frac{\cancel{1}}{\cancel{\theta}} \cdot \frac{e^{-y/\theta}}{\frac{\cancel{1}}{\cancel{\theta}} \cdot -1} = \left[-e^{-y/\theta}\right]_0^{u^{1/a}}$$

$$= -e^{\frac{-u^{1/a}}{\theta}} - (-e^0) = 1 - e^{\frac{-u^{1/a}}{\theta}}, \quad u > 0$$

$$f_U(u) = \frac{d}{du}\, 1 - e^{\frac{-u^{1/a}}{\theta}} = 0 - e^{\frac{-u^{1/a}}{\theta}} \cdot \frac{1}{\theta} \cdot \frac{1}{a} \cdot -u^{1/a - 1}$$

$$\boxed{= \frac{u^{1/a - 1} \cdot e^{\frac{-u^{1/a}}{\theta}}}{a\theta}}$$

Example B : If Y is a continuous random variable with density function

$$f_Y(y) = \begin{cases} \dfrac{3y^2}{16} & , \quad -2 < y < 2 \\ 0 & , \quad else \end{cases},$$

what is the density function of $U = 1 - 6Y^2$?

$$U = 1 - 6Y^2$$

$$P(U < u) = P(1 - 6Y^2 < u) = P(1 - u < 6Y^2) = P\left(\frac{1-u}{6} < Y^2\right)$$

$$= P\left(Y > \pm\sqrt{\frac{1-u}{6}}\right) = P\left(Y > \sqrt{\frac{1-u}{6}}\right) + P\left(Y < -\sqrt{\frac{1-u}{6}}\right)$$

$$= \int_{\sqrt{\frac{1-u}{6}}}^{\infty} \frac{3y^2}{16}\,dy + \int_{-\infty}^{-\sqrt{\frac{1-u}{6}}} \frac{3y^2}{16}\,dy = \int_{\sqrt{\frac{1-u}{6}}}^{2} \frac{3y^2}{16}\,dy + \int_{-2}^{-\sqrt{\frac{1-u}{6}}} \frac{3y^2}{16}\,dy$$

$$= \left[\frac{\cancel{3}}{16} \cdot \frac{y^3}{\cancel{3}}\right]_{\sqrt{\frac{1-u}{6}}} + \left[\frac{\cancel{3}}{16}\,\frac{y^3}{\cancel{3}}\right]_{-2}^{-\sqrt{\frac{1-u}{6}}}$$

$$= \left[\frac{y^3}{16}\right]_{\sqrt{\frac{1-u}{6}}}^{2} + \left[\frac{y^3}{16}\right]_{-2}^{-\sqrt{\frac{1-u}{6}}}$$

$$\frac{2^3}{16} - \frac{\left(\sqrt{\frac{1-u}{6}}\right)^3}{16} + \frac{\left(-\sqrt{\frac{1-u}{6}}\right)^3}{16} - \frac{(-2)^3}{16}$$

$$\frac{8}{16} - \frac{\left(\sqrt{\frac{1-u}{6}}\right)^3}{16} - \frac{\left(\sqrt{\frac{1-u}{6}}\right)^3}{16} + \frac{8}{16}$$

$$= 1 - \frac{2\left(\sqrt{\frac{1-u}{6}}\right)^3}{16} = 1 - \frac{\left(\sqrt{\frac{1-u}{6}}\right)^3}{8}$$

$$f_U(u) = \begin{cases} 1 - \frac{1}{8} \cdot \left(\sqrt{\frac{1-u}{6}}\right)^3, & -23 < u < 1 \\ 0, & \text{otherwise.} \end{cases}$$

Example C:

The life (in hundreds of hours) of a type of battery running a particular machine follows an *Exponential* distribution with mean 1. All batteries are independent. Let X denote the life of battery 1, and Y denote the life of battery 2. $E[X]=1$, $X \sim Exp(\beta=1)$ $Y \sim Exp(\beta=1)$

a) What is the probability density of the average life of two of these batteries?

$$f_X(x) = \frac{1}{1} \cdot e^{-x/1}, \quad x>0$$

$$f_Y(y) = \frac{1}{1} e^{-y/1}, \quad y>0$$

$$f(x,y) = f(x) \cdot f(y) = e^{-(x+y)}, \quad x>0, y>0$$

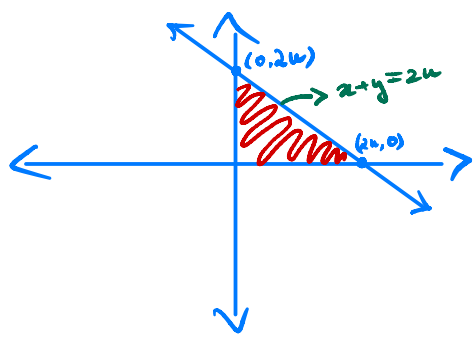
$$U = \frac{X+Y}{2}$$

$$P(U<u) = P\left(\frac{X+Y}{2} < u\right) = P(X+Y < 2u)$$

Case 1

$2u > 0$

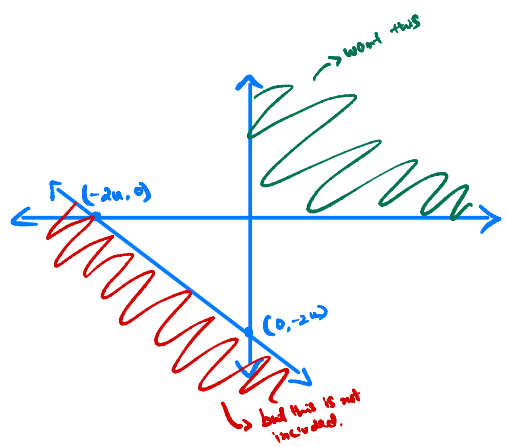
$$\int_0^{2u-x} \int f(x,y)\, dy\, dx$$



Case 2

$2u < 0$

∴ do not need to consider.



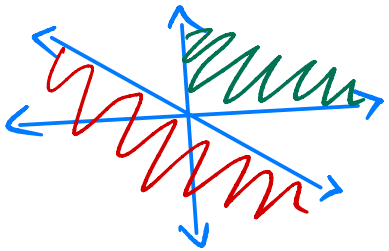
b) Consider two batteries. What is the probability density for the proportion of the total time the machine ran off battery 1? That is, what is the distribution of $U = \dfrac{X}{X+Y}$?

$$P(U<u) = P\left(\frac{X}{X+Y} < u\right) = P(X < u(X+Y))$$

$$= P(0 < uX + uY - X) = P(0 < X(u-1) + uY) = P(-uY < X(u-1)) = P\left(Y > \left(\frac{1-u}{u}\right)X\right)$$

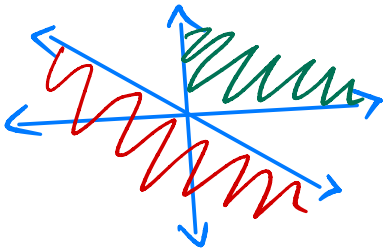
Case 1

$u > 1$, $\dfrac{1-u}{u}$ = -ve Slope



Case 2

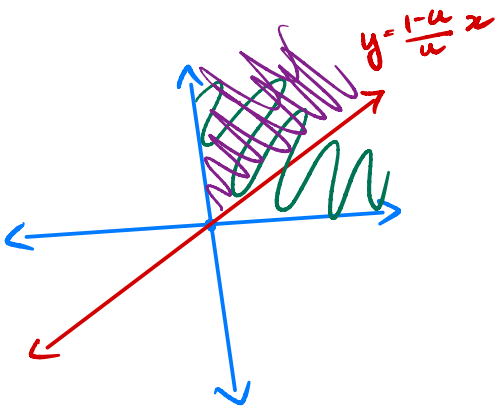
$u < 0$, $\dfrac{1-u}{u}$ = -ve Slope



Case 3

$0 < u < 1$, $\dfrac{1-u}{u}$ = +ve Slope

$$\int_0^\infty \int_{\frac{1-u}{u}x}^\infty e^{-(x+y)}\,dy\,dx$$



Example D : The joint density of X and Y is given as

$$f(x,y)=\begin{cases}\frac{1}{2} & , \quad 0<y<2, 0<x<1, \\ 0 & , \quad else\end{cases}$$

Find the probability density function of $V=Y-X$.